

Vishesh Gupta

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SKILLS

Languages // Python, Go, C/C++, JavaScript/TypeScript

Frameworks & Libraries // gRPC, GraphQL, REST, React, Hugging Face, Lang Chain, NumPy, Pandas,

Tools // Docker, Kubernetes, AWS, GCP, Git, GitHub Actions, PostgreSQL, Protocol Buffers, Kafka Redis, Redpanda, RabbitMQ, IBM MQ

EXPERIENCE

SENIOR SOFTWARE ENGINEER // ALPACA MARKETS

GLOBALLY REMOTE

MAR 2025 – PRESENT

- Became OCC Infra SME in under 3 weeks, leading critical self-clearing integration under tight deadlines and adding resiliency & DR
- Owned network interconnect architecture and vendor coordination for the core trading business, ensuring low-latency, fault-tolerant connectivity with robust disaster recovery capabilities and enhanced observability to receive network level alerts from vendors
- Automated previously manual, communication-heavy processes into self-serve tooling, greatly improving developer velocity and QoL
- Mentored engineering team members and established best practices to enhance code quality, collaboration, and operational excellence
- Built Terraform modules for auto-clustered environments, cutting region launch time by 80% and enabling rapid, scalable deployments

SOFTWARE ENGINEER – INFRA LEAD // MONOCEROS

GRAND CAYMAN, KY

MAY 2024 – JAN 2025

- Independently built and managed Software and Hardware Infrastructure for 4 teams to handle market data flowing in the system
- Reliable infrastructure supported trading operations for multi-million dollars' worth of trading making huge impacts to all our traders
- Handled physical hardware for networking in multi office setup that ensures hardware reliability with best practices for high uptime
- Improved the developer experience by creating state of the art Infrastructure tooling to deploy systems, sandboxed environments, etc.
- Managed vendor communication and partner interactions to ensure rapid issue resolution and maintain critical system access

SOFTWARE ENGINEER // CLEAR STREET

NEW YORK, NY (REMOTE)

JAN 2021 – APR 2024

- Orchestrated integrations for Vault, Teleport, Artifactory, and Buf across engineering team, enforcing best practices and improving security and workflow efficiency
- Generated IaC automation tooling for Infrastructure Resource provisioning and faster CI/CD system with proper reviews and permission system. Modules to simplify resource creation in the IaC, reducing overhead by 40% in access request creation.
- Improved internal DNS routing using Transit Gateway and Route 53 Resolvers to handle DNS issues over the internal VPN network

PROJECTS

CRYPTOGRAPHIC ALGORITHMS // Basic encryption and decryption techniques

- Programmed basic concepts from cryptography such as Symmetric Key, Hashing, and Block Cipher
- Gained a strong understanding of Private and Public Key Cryptography and security techniques that help write secure software.

STATISTICAL MODELS & DISTRIBUTIONS // Collection of statistical models found in random research papers or articles

- Container to statistical models built from University courses or by reading about them in some research paper using R and Python
- Covers different distribution sets and graphical representations of datasets using Tableau and R libraries such as ggplot2

EDUCATION

UNDERGRADUATE // UNIVERSITY OF WATERLOO, WATERLOO, ONTARIO

SEPT 2016 – JAN 2021

Bachelor of Computer Science, Honors, Co-op // East Asian Studies Minor

Relevant Coursework: Data Structures & Algorithm, Operating Systems (C), Object Oriented Programming (C++), Numerical Computation (Python/Jupyter), Applied Cryptography

HOBBIES & INTERESTS

SPORTS & ADVENTURE ACTIVITIES // Badminton, Volleyball, Basketball, Baseball, Rock Climbing, Hiking, Skiing

INTERESTS // Anime, Music, Fine Dining, Networking, Travelling